Sustainability Economics: Complete Lecture Notes

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Course Overview & Syllabus Summary

What is the connection between economics and sustainability? When people talk about economics, often they are talking about something to do with money. How people acquire it and what they do with it. When people talk about sustainability, often they are talking about the environment, how it is changing, and whether or not those changes are good. Where do these two things come together? That intersection often seems to be about the way that people are acquiring and using money is changing the environment in ways that are not good.

Part I

Foundations & Basic Principles

Chapter 1

Introduction & Key Concepts

1.1 Overview & Objectives

The term *sustainability* implies continuity and balance. If something is sustainable, it can continue doing what it is doing. Unless it is doing nothing, what it is doing is interacting with other entities. This implies that, except for those things that are doing nothing, we should think about sustainability with a systems perspective.

A system is a collection of entities (or elements) and the connections between them. To understand a system, we need to understand the entities and their connections. Our goal in this course is to develop an understanding of the relationship between the economy and the environment by looking at the elements that make up the economy and their relationship with elements that make up the environment. To do so, we will proceed by building a model of a *complex adaptive social ecological system*.

The system we inhabit, and that we will try to understand, is complex. This means that the system as a whole exhibits properties that cannot be explained by looking at any element itself. It is adaptive in that it changes in response to internal and external pressures.

ey reading for this section: Chapter 1 of the required textbook.

Systems

- A system is a collection of elements and their connections.
- A social ecological system contains elements in both the social (human) and ecological (nature) domains, and has connections crossing between the domains.
- A system is complex when there are 'emergent' properties that cannot be predicted looking at elements in isolation.
- A system is adaptive if it continuously evolves or changes in response to internal and external forces.

Systems



1.1.1 Detailed Explanation of Concept 1

While the slide highlights the main points, it is important to note that the theoretical model relies on two primary assumptions...

Mathematical Representation

$$f(x) = \alpha + \beta x + \varepsilon$$

Chapter 2

Systems

2.1 Systems

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Systems



When we think about systems, we need to consider scale. If we consider the scale of ...

Scales

- What constitutes elements of a system depends on the physical scale at which we are contemplating the system.
  - At the atomic scale, atoms are the elements.
  - At the cosmic scale, galaxies are the elements.
  - What are the elements at the human scale?
- What the connections between elements are depends on the physical scale and the temporal scale chosen.
  - How long does it take for the connection to change something for the elements that we are interested in?
  - For the movement of atoms ...
  - For the movement of galaxies ...
  - For the human issues we are concerned about?

In both the natural and social systems, researchers try to identify patterns in the systems that they are studying. Those patterns emerge from the interactions between the elements that make up the system. When systems are large and complex, the discipline typically finds ways to describe the patterns without examining the detail of the interacting entities that leads to the pattern.

Cosmologists studying the movement of galaxies do not spend much time considering the movement of individual hydrogen atoms. However, it is the movement of those individual hydrogen atoms, under the influence of gravity, that leads to the formation of stars, and that leads to stars spinning together in galaxies.

Often, the disciplines have observed or thought deeply about the entities that they are interested in, and arrived at relationships - often mathematical - that represent the patterns that they observe.

In chemistry, the relationship $PV = nRT$ is known at the Ideal Gas Law. It is the relationship between pressure (P), volume (V) and temperature (T). The amount of gas in the volume is (n) in units of moles, and R is the ideal gas constant, expressed the the appropriate units. A mole is Avagadro's number of atoms. This relationship expresses how temperature, which is a measure of how much energy atoms have, and the pressure that they are exerting within a particular volume. The physics of the situation is that the atoms are racing around, their speed reflecting the temperature, and smashing into the sides of the container that confines them. The more energy they have, the faster they are moving, and the harder they collide with the chamber. However, the equation is not representing the movement of those individual atoms. Rather, it is expressing this effect in a way that is relevant for things like the operation of a steam engine.

In the last few decades, computers have become powerful enough for us to represent the details of relationships like this from the basic building blocks. Agent based models are a class of such tools. One such agent based modeling environment we will use from time to time is NetLogo (https://www.netlogo.org/).

Agent Based Models

- Agent Based Models (ABMs) that interact virtual agents in virtual environments, ot observe how the system evolves.
- Agents can range from individual atoms to galaxies.

When people who are concerned about the environment think of economics, they often conclude that it ignores the environment. The way introductory economics is taught often reinforces this, as if the environment is mentioned at all, it is a small topic, externalities.

There is a practical reason for this. During a presidential debate, former US president Bill Clinton said "It's the economy, stupid." [CONFIRM QUOTE]. He was emphasizing that what people are most worried about is their own situation. In a modern economy, your situation depends on your income and what it costs you to live. The environment is important, but is secondary to being able to put food on the table tonight, and having a roof under which that table is placed.

You are in this course because you are concerned about the environment, about sustainability. Therefore, lets start by building our understanding of sustainability economics from that environment.

The Fish Pond

- The Question: How does the population of fish in a pond change over time?
  - What time step?

  - One minute? - population changes small, unless major event.
  - One century? - population change in relation to long term events.
  - One year? - scale that matches seasons.

- The elements of a fish pond.
  - The fish (each fish, or fish as a population - biomass?).
  - The water (each molecule, or as a medium for the fish?).
  - The food (each food item - e.g phytoplankton, insects, other fish - or as a population - biomass?).

- The connections within the fish pond system.
  - Fish encountering food and eating.
  - Fish health and survival affected by state of the water.
  - Fish encounter each other and reproduce.

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Fish example: https://virtualbiologylab.org/ModelsHTML5/PopGenFishbowl/PopGenFishbowl.html

Chapter 3

Hermit Economics

3.1 Learning Objectives

- Explain the meaning of opportunity cost.
- Demonstrate an understanding of the production function and the production possibilities frontier.
- Explain the role of diminishing marginal product.

3.2 The Hermit

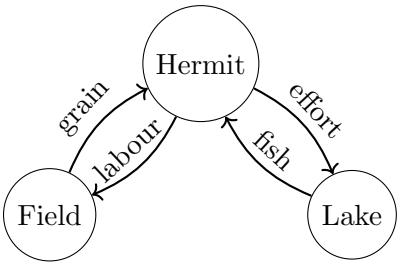
A hermit is someone who has chosen to live apart from society. In the most extreme case, they have no contact with any other person and are completely self sufficient. From a system thinking perspective, the hermit would be one element of the system, with the other elements being part of the ecological side of the hermit-nature social ecological system.

Hermit System

- Hermit,
  - The social element of the social ecological system.
  - Contributes labour and consumes the fruits of that labour.
- Field,
  - One ecological element of the system.
  - Produces grain, with labour as the main input.
- Lake.
  - One ecological element of the system.
  - Produces fish, with fishing effort (time spent fishing) as the main input.

The hermit can apply their time either to labour in the field, growing grain, or effort at the lake to catch fish. The more labour applied in the field, the more grain that the hermit harvests. Likewise, the more effort devoted to fishing, the more fish that will be caught.

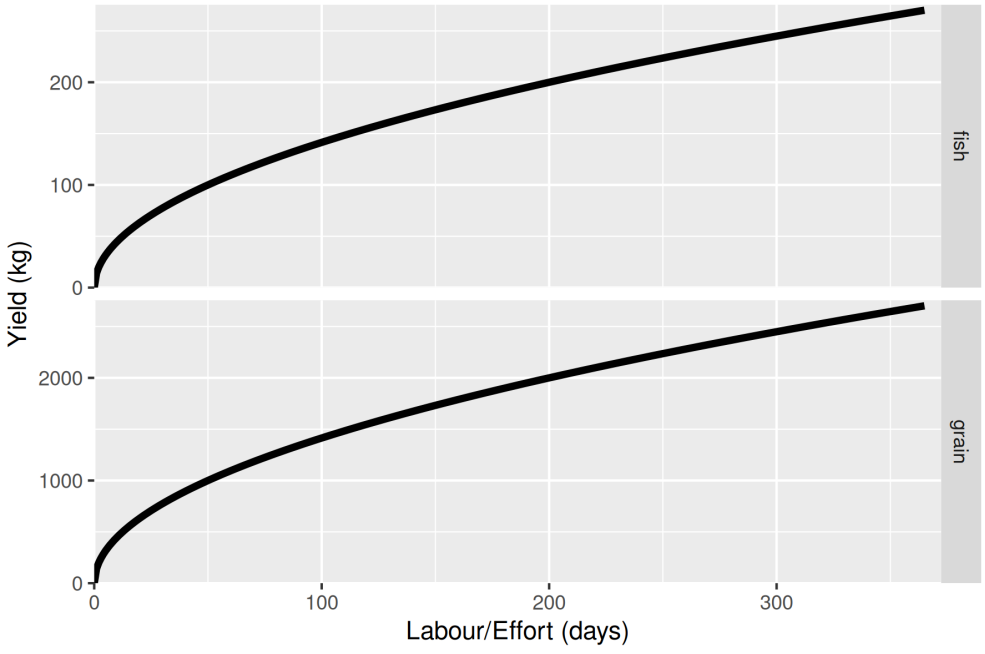
Hermit System



Production Functions

- A *technology* is a way that inputs are converted to outputs.
- A *production function* is a way of representing the technology.
  - The production function is a simplification.
  - The technology behaves the way that physics, chemistry, etc. require it to.
  - All models are wrong, some are useful.
- Three measures are commonly used to describe production at each input level.
  - *Total Product* is the total output produced given a particular input level.
  - *Average Product* is the total product divided by the input level.
  - *Marginal Product* is how much the total output changes if the level of input is changed by a small amount.

Production Functions



Total, Average and Marginal Product

- Total Product (TP) and Average Product (AP) are based on a level of input.

$$AP = TP/L$$

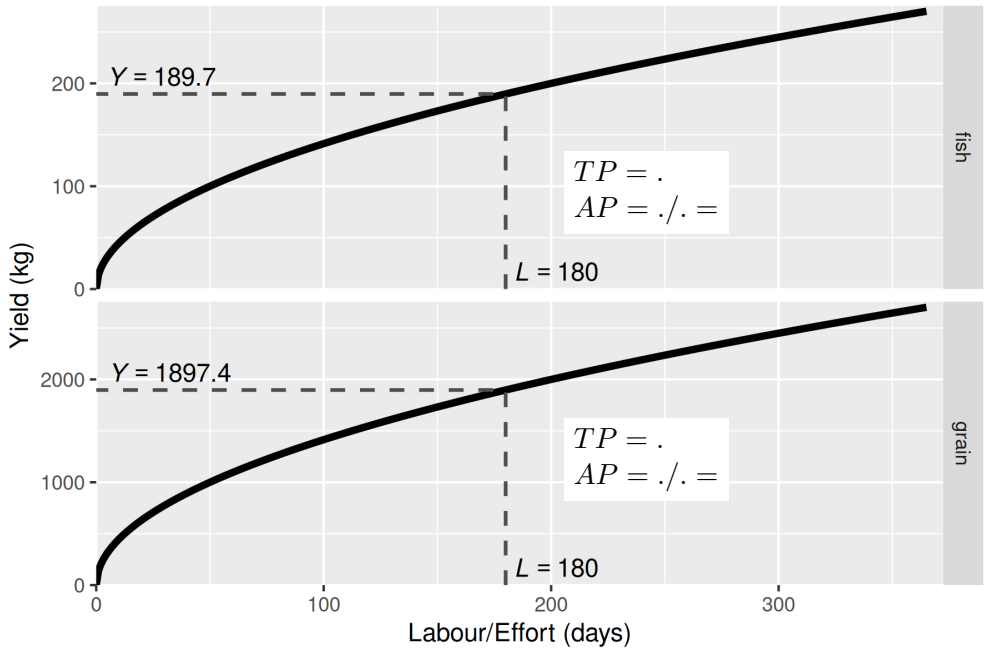
  - L can be any input, not just labour.

- Marginal Product (MP) is based on a small change in in the level of an input, around a specific level of that input.

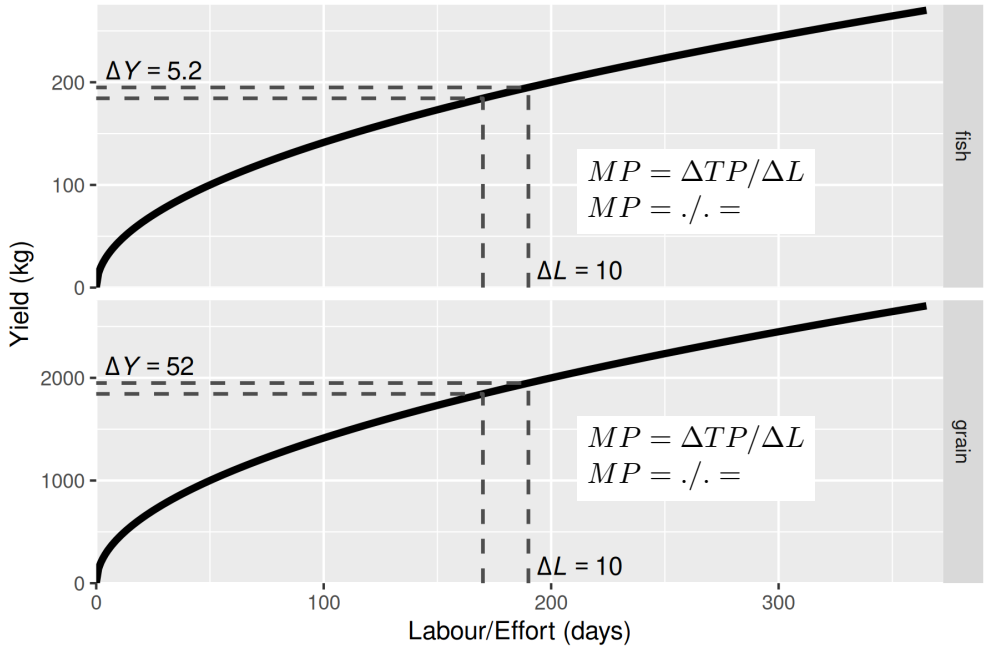
$$MP = \Delta TP/\Delta L = \partial TP/\partial L$$

  - The marginal product measures how much extra output would be produced if a small amount of extra input is used.
  - It is the slope of a line that just touches the production function at the input level.

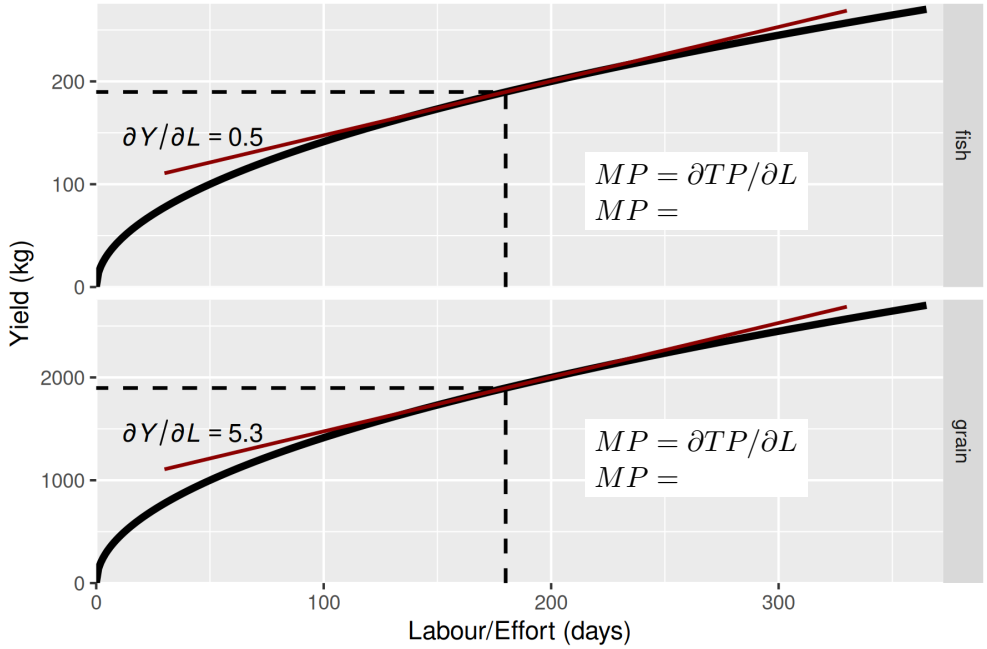
Total, Average and Marginal Product



Total, Average and Marginal Product



Total, Average and Marginal Product

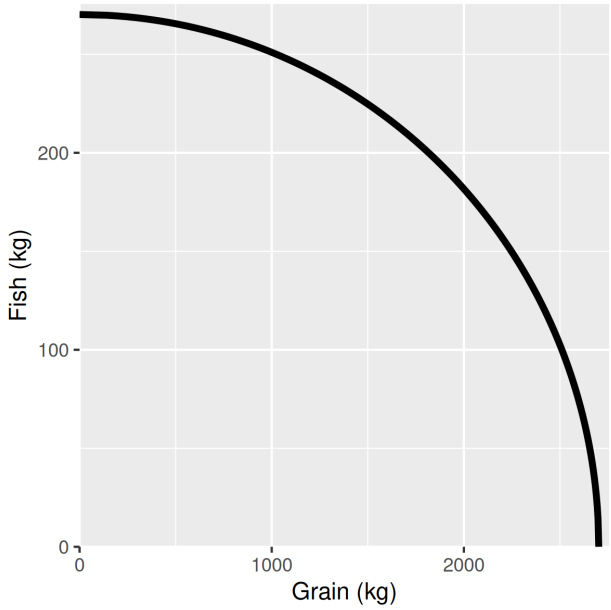


Production Possibilities Frontier

- The hermit has a maximum of 365 days per year to either work in the grain field or catch fish.
- Spending more time working in the grain field means less time catching fish.
- Spending less total time on both activities means less fish, less grain, or less of both.

- How many days of the year should the hermit spend on these two activities?

Production Possibilities Frontier



Opportunity Cost

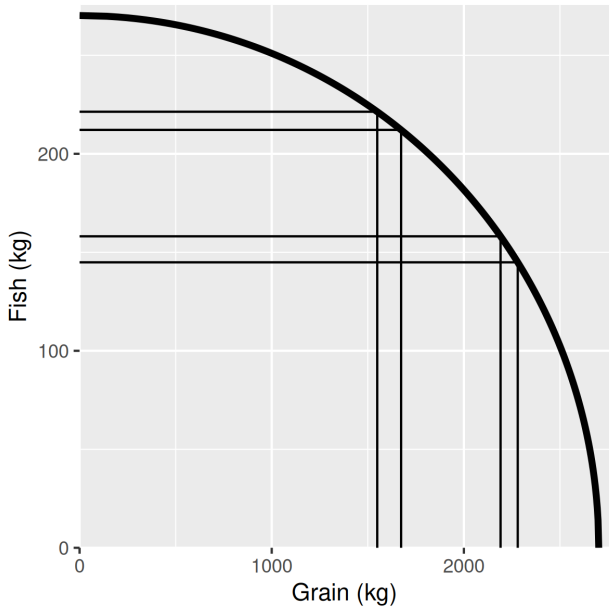
- The hermit has a maximum of 365 days per year to either work in the grain field or catch fish.

- Spending more time working in the grain field means less time catching fish.

- The *opportunity cost* of the grain produced by spending one more day in the field is the fish that would have been caught if that day had been spent catching fish.

- The ***opportunity cost*** of a choice is the value of the next best alternative foregone.

Production Possibilities Frontier



3.2.1 Detailed Explanation of Concept 1

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